DAY 26 | POST-READ

Orchestration, Tools and LangGraph: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Chains (LCEL)	prompt | model | parser; invoke one input, batch many.
LLM as judge	A second call scores an answer with a rubric. Calibrate before you trust it.
create_agent (NEW v4)	LangChain v1's one-call tool-calling agent, built on LangGraph.
Middleware (NEW v4)	Hooks before and after model and tool calls: limits, approvals, PII, logging.
Tool calling	Model proposes; code validates (Pydantic), allowlists and executes.
Tool token cost (NEW v4)	Schemas and results are input tokens on every call. Keep tools few and lean.
Agents SDK (NEW v4)	Agent, function_tool, Runner, handoffs, guardrails, tracing.
LangGraph	State, nodes, edges, conditional edges, cycles.
Durable execution (NEW v4)	Checkpointer per thread_id; resume after failure; interrupt() and Command(resume).

Part B: Going beyond today

1. Tool search

When an agent has dozens of tools, send only the relevant ones. LangChain offers a tool selector middleware and provider tool search. Fewer tools per call means fewer tokens and better choices.

2. Parallel tool calls

Models can request several independent tool calls in one turn, for example two order lookups. Run them in parallel, but keep risky tools sequential and approved.

3. Production checkpointers

InMemorySaver is for demos. In production use a database-backed checkpointer so a restart does not lose paused approvals. Clean up old threads on a schedule.

4. Streaming

Agents can stream tokens and intermediate steps to the UI so users see progress. Stream the path trace as well as the answer.

5. Version everything

Pin langchain, langgraph and openai-agents versions in requirements. Agent frameworks change quickly; upgrade on purpose with tests (Day 33, AgentOps).

Part C: Check yourself

- Why does the model never execute a tool itself?
- Name three middleware hooks and one use for each.
- How do tool definitions affect the token bill?
- What happens when you invoke a paused graph with Command(resume=...)?
- When would you choose the direct API over a framework?

Part D: Practice ideas for tonight

- Add a search_policy tool to the create_agent assistant and measure the extra schema tokens.
- Make the refund approval interrupt reject by default and log every decision.
- Print the Mermaid diagram of your graph and paste it into your SmartDesk README.

Coming up on Day 27: agentic AI, corrective RAG, multi-agent systems, persistent memory and GenAI assessment 2.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today